

NETWORKS AND COMPLEXITY

Solution 10-10

*This is an example solution from the forthcoming book *Networks and Complexity*.*

Find more exercises at <https://github.com/NC-Book/NCB>

Ex 10.10: Viral attacks in general [Lvl. 3]

Now, we derive some results on the viral attack. Let G be the degree generating function before the attack, and y the proportion of links that are not part of the giant conducting component.

- a) Show that a viral attack starting within the giant conducting component is described by the attack generating function

$$R = G - G(xy)$$

Note that we have omitted the argument (x) behind R and the first G , in accordance with our notation.

Solution

From the chapter we know that the probability that a randomly picked node has degree k and is subsequently removed in the viral attack is

$$r_k = p_k(1 - y^k). \quad (1)$$

Furthermore we know the definition of the attack function

$$R = \sum r_k x^k. \quad (2)$$

Substituting r_k , we can write

$$R = \sum p_k(1 - y^k)x^k \quad (3)$$

$$= \left(\sum k p_k x^k \right) - \left(\sum p_k (xy)^k \right) \quad (4)$$

$$= G(x) - G(xy) = G - G(xy) \quad (5)$$

where we used $G(x) = \sum p_k x^k$ in the last step.

- b) Now start from $R = G - G(xy)$ and show that the properties of the network after the viral attack are given by

$$N_a = G(y)N,$$

$$z_a = y^2 \frac{Q(y)}{G(y)} G'(y), \quad q_a = y^2 Q'(y),$$

$$G_a = \frac{G(\tilde{A}y)}{G(y)}, \quad Q_a = \frac{Q(\tilde{A}y)}{Q(y)},$$

where $\tilde{A} = \tilde{c}x + \tilde{r}$ and $\tilde{c} = 1 - \tilde{r} = yG'(y)/z = yQ(y)$.

Solution

Number of nodes. To find the equation for N_a we need to work out what proportion of the network is removed in the attack:

$$r = R(1) = G(1) - G(y) = 1 - G(y), \quad (6)$$

where we used $G(1) = 1$. The surviving fraction of the nodes is

$$c = 1 - r = G(y). \quad (7)$$

Multiplying this surviving fraction with the total number of nodes gives us the desired result

$$N_a = cN = G(y)N. \quad (8)$$

Mean degree. To find the result we use the respective result for degree targeted attacks

$$z_a = \frac{z\tilde{c}^2}{c}. \quad (9)$$

In our calculation of the surviving number of nodes we have just seen $c = G(y)$ and from our analysis of the degree-targeted attack we know

$$\tilde{r} = \frac{R'(1)}{z}. \quad (10)$$

Substituting $R = G - G(xy)$ yields

$$\tilde{r} = \frac{G'(1) - yG'(y)}{z} \quad (11)$$

$$= \frac{z - yG'(y)}{z} \quad (12)$$

$$= 1 - \frac{yG'(y)}{z} \quad (13)$$

$$= 1 - yQ(y) \quad (14)$$

where the factor y appears as the inner derivative from differentiating $G(xy)$ with respect to x . We can now find the $\tilde{c}$ as the complement of $\tilde{r}$,

$$\tilde{c} = 1 - \tilde{r} = yQ(y). \quad (15)$$

This gives us all the parts that we need to put the solution together, we start the equation for z_a and substitute c and $\tilde{c}$,

$$z_a = \frac{z\tilde{c}^2}{c} \quad (16)$$

$$= \frac{z(yQ(y))^2}{G(y)} \quad (17)$$

$$= y^2 \frac{Q(y)}{G(y)} zQ(y) \quad (18)$$

$$= y^2 \frac{Q(y)}{G(y)} G'(y) \quad (19)$$

where we have used $Q = G'/z$ in the last step to make the result a little bit nicer. This form is appealing because $z = G'(1)$, so having a G' appear on the right hand side is neat.

An alternative way is to first derive

$$G_a = \frac{G(\tilde{A}y)}{G(y)} \quad (20)$$

and then compute z_a using the relationship between degree generating function and the mean degree

$$z_a = G'_a(1) \quad (21)$$

In this case we must take care with the derivative as we are differentiating with respect to x , which appears as the argument of $\tilde{A}$, (remember $\tilde{A} = \tilde{A}(x)$). We find

$$z_a = \left. \frac{\partial}{\partial x} \frac{G(\tilde{A}y)}{G(y)} \right|_{x=1} \quad (22)$$

$$= \left. \frac{G'(\tilde{A}y)\tilde{A}'y}{G(y)} \right|_{x=1} \quad (23)$$

$$= \frac{G'(\tilde{A}(1)y)\tilde{A}'(1)y}{G(y)} \quad (24)$$

$$= \frac{G'(y)\tilde{A}'(1)y}{G(y)} \quad (25)$$

$$= \frac{G'(y)\tilde{c}y}{G(y)} \quad (26)$$

$$= \frac{G'(y)y^2Q(y)}{G(y)} \quad (27)$$

$$= y^2 \frac{Q(y)}{G(y)} G'(y) \quad (28)$$

where we used $\tilde{A}(1) = 1$ and $\tilde{A}'(1) = \tilde{c}$ and $\tilde{c} = yQ(y)$, derived above.

Excess degree. As for the mean degree there are two ways to reach the desired result. First, we can use the equation for the excess degree after a degree targeted attack $q_a = q - R''(1)/z$. We start by computing the second derivative of R ,

$$R'' = \frac{\partial^2}{\partial x^2} (G - G(xy)) \quad (29)$$

$$= G'' - y^2 G'''(xy) \quad (30)$$

Substituting into the equation for q_a yields

$$q_a = q - \frac{R''(1)}{z} \quad (31)$$

$$= q - \frac{G''(1) - y^2 G'''(y)}{z} \quad (32)$$

$$= q - \frac{G''(1)}{z} + \frac{y^2 G'''(y)}{z} \quad (33)$$

$$= q - q + \frac{y^2 G'''(y)}{z} \quad (34)$$

$$= \frac{y^2 G'''(y)}{z} \quad (35)$$

$$= y^2 Q'(y) \quad (36)$$

where we used $Q' = G''/z$ in the last step, which follows from $Q = G'/z$.

Alternatively, we can first derive the equation for the excess degree generating function $Q_a = Q(\tilde{A}y)/Q(y)$ and then compute the mean excess degree in the usual way,

$$q_a = Q'_a(1) \quad (37)$$

$$= \left. \frac{\partial}{\partial x} \frac{Q(\tilde{A}y)}{Q(y)} \right|_{x=1} \quad (38)$$

$$= \left. \frac{Q'(\tilde{A}y)\tilde{A}'y}{Q(y)} \right|_{x=1} \quad (39)$$

$$= \frac{Q'(\tilde{A}(1)y)\tilde{A}'(1)y}{Q(y)} \quad (40)$$

$$= \frac{Q'(y)\tilde{c}y}{Q(y)} \quad (41)$$

$$= \frac{Q'(y)Q(y)y^2}{Q(y)} \quad (42)$$

$$= y^2 Q'(y) \quad (43)$$

which is again the desired result.

Degree generating function. Here we use the result from degree-targeted attacks, which tells us

$$G_a = \frac{G(\tilde{A}) - R(\tilde{A})}{c} \quad (44)$$

$$= \frac{G(\tilde{A}) - (G(\tilde{A}) - G(\tilde{A}y))}{c} \quad (45)$$

$$= \frac{G(\tilde{A}y)}{c} \quad (46)$$

$$= \frac{G(\tilde{A}y)}{G(y)}. \quad (47)$$

Excess degree generating function. Here we have again the choice between two different ways. Using the result from degree-targeted attacks we can write

$$Q_a = \frac{G'(\tilde{A}) - R'(\tilde{A})}{z\tilde{c}} \quad (48)$$

$$= \frac{G'(\tilde{A}) - (G'(\tilde{A}) - G'(\tilde{A}y))}{z\tilde{c}} \quad (49)$$

$$= \frac{yG'(\tilde{A}y)}{z\tilde{c}} \quad (50)$$

$$= \frac{yG'(\tilde{A}y)}{z} \frac{z}{yG'(y)} \quad (51)$$

$$= \frac{G'(\tilde{A}y)}{G'(y)}. \quad (52)$$

To bring this into the desired form we need to expand the fraction by $1/z$ in order to use $Q = G'/z$, which gives us

$$Q_a = \frac{G'(\tilde{A}y)}{G'(y)} \quad (53)$$

$$= \frac{G'(\tilde{A}y)/z}{G'(y)/z} \quad (54)$$

$$= \frac{Q(\tilde{A}y)}{Q(y)} \quad (55)$$

which is what we wanted.

The alternative way to construct the excess degree generating function after the attack is to compute it from the degree generating function using $Q = G'/z$ applied to the functions after the attack, so

$$Q_a = \frac{G'_a}{z_a} = \frac{G'_a}{G'_a(1)} \quad (56)$$

To make use of this relationship, lets start by computing

$$G'_a = \frac{\partial}{\partial x} \frac{G(\tilde{A}y)}{G(y)} \quad (57)$$

$$= \frac{G'(\tilde{A}y)\tilde{A}y}{G(y)} \quad (58)$$

$$= \frac{G'(\tilde{A}y)\tilde{A}'y}{G(y)} \quad (59)$$

$$= \frac{G'(\tilde{A}y)\tilde{c}y}{G(y)} \quad (60)$$

It is now tempting to substitute the equation for $\tilde{c}$ that we derived above, but in this case it is unnecessary because this factor is about to get cancelled anyway as we substitute into the equation for Q_a . Cancel culture, here we come,

$$Q_a = \frac{G'_a}{G'_a(1)} \quad (61)$$

$$= \frac{G'(\tilde{A}y)\tilde{c}y}{G(y)} \frac{G(y)}{G'(\tilde{A}(1)y)\tilde{c}y} \quad (62)$$

$$= \frac{G'(\tilde{A}y)}{G'(\tilde{A}(1)y)} \quad (63)$$

$$= \frac{G'(\tilde{A}y)}{G'(y)} \quad (64)$$

$$= \frac{G'(\tilde{A}y)/z}{G'(y)/z} \quad (65)$$

$$= \frac{Q(\tilde{A}y)}{Q(y)} \quad (66)$$